Polar Curves and Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting between polar and rectangular coordinates**

Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.

Multiply by r: $r^2 = 4r\cos\theta \Rightarrow x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0 \Rightarrow (x-2)^2 + y^2 = 4$. This is a circle of radius 2 centered at $(2, 0)$.

2 **[Foundational]** • **Target: Converting between polar and rectangular coordinates**

Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.

$x = r\cos\theta = 4\cos\left(\frac{\pi}{3}\right) = 4\left(\frac{1}{2}\right) = 2$. $y = r\sin\theta = 4\sin\left(\frac{\pi}{3}\right) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$. Point: $(2, 2\sqrt{3})$.

3 **[Foundational]** • **Target: Identifying and sketching common polar curves**

Identify the type of curve for $r = 2 + 2\sin\theta$, and state its key feature.

This is a cardioid (since the coefficient of $\sin\theta$ equals the constant term) — a heart-shaped curve passing through the pole.

4 **[Foundational]** • **Target: Identifying and sketching common polar curves**

Identify the type of curve for $r = \cos(3\theta)$.

This is a rose curve with 3 petals (odd n in $r = \cos(n\theta)$ gives n petals).

5 **[Foundational]** • **Target: The slope of a polar curve**

State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos\theta$ and $y = r\sin\theta$.

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$.

6 **[Foundational]** • **Target: The slope of a polar curve**

Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.

$\frac{dr}{d\theta} = 2$. $x = 2\theta\cos\theta$, $y = 2\theta\sin\theta$. $\frac{dy}{d\theta} = 2\sin\theta + 2\theta\cos\theta$; $\frac{dx}{d\theta} = 2\cos\theta - 2\theta\sin\theta$. At $\theta = \frac{\pi}{2}$: $\frac{dy}{d\theta} = 2(1) + 2(\frac{\pi}{2})(0) = 2$; $\frac{dx}{d\theta} = 2(0) - 2(\frac{\pi}{2})(1) = -\pi$. $\frac{dy}{dx} = \frac{2}{-\pi} = -\frac{2}{\pi}$.

7 **[Foundational]** • **Target: Finding horizontal and vertical tangents**

For the circle $r = 4\cos\theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin\theta$ has a critical point).

$y = 4\cos\theta\sin\theta = 2\sin(2\theta)$. $\frac{dy}{d\theta} = 4\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Symmetry of polar curves**

Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x-axis).

Replace θ with $-\theta$ in the equation. If the equation is unchanged (or yields the same curve), the curve is symmetric about the polar axis. For example, $r = 2 + 2\cos\theta$ satisfies $f(-\theta) = 2 + 2\cos(-\theta) = 2 + 2\cos\theta = f(\theta)$, confirming symmetry.

9 **[Intermediate]** • **Target: Symmetry of polar curves**

Test whether $r = 3\sin\theta$ is symmetric about the y-axis (replace θ with $\pi - \theta$).

$f(\pi - \theta) = 3\sin(\pi - \theta) = 3\sin\theta = f(\theta)$ (using $\sin(\pi - \theta) = \sin\theta$). Symmetric about the y-axis.

10 **[Intermediate]** • **Target: More polar-to-rectangular conversions**

Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \le \theta < 2\pi$).

$r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$. $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ (in Quadrant I, matching both coordinates positive).

11 **[Intermediate]** • **Target: More polar-to-rectangular conversions**

Convert the rectangular equation $y = x$ to polar form.

$r\sin\theta = r\cos\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ (a constant-θ ray, i.e. a straight line through the pole).

12 **[Intermediate]** • **Target: More polar-to-rectangular conversions**

Convert the polar equation $r\sin\theta = 4$ to rectangular form, and identify the curve.

$r\sin\theta = y$, so $y = 4$: a horizontal line.

13 **[Intermediate]** • **Target: More slope-of-a-polar-curve practice**

Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.

$\frac{dr}{d\theta} = \cos\theta$. $\frac{dy}{dx} = \frac{\cos\theta\sin\theta + r\cos\theta}{\cos\theta\cos\theta - r\sin\theta}$. At $\theta = 0$: $r = 1$, $\frac{dr}{d\theta} = 1$. Numerator: $1(0) + 1(1) = 1$. Denominator: $1(1) - 1(0) = 1$. $\frac{dy}{dx} = \frac{1}{1} = 1$.

14 **[Intermediate]** • **Target: More slope-of-a-polar-curve practice**

Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ, and interpret the result using the fact that a circle centered at the origin has radial symmetry.

$\frac{dr}{d\theta} = 0$. $\frac{dy}{dx} = \frac{0 \cdot \sin\theta + 5\cos\theta}{0 \cdot \cos\theta - 5\sin\theta} = \frac{5\cos\theta}{-5\sin\theta} = -\cot\theta$. This matches the slope of the tangent line to a circle at angle θ — perpendicular to the radius at that point, consistent with basic circle geometry.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the values of θ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

$1 - \cos\theta = 0 \Rightarrow \cos\theta = 1 \Rightarrow \theta = 0$. At the single angle where $r = 0$, the curve touches the pole; the tangent line there is along $\theta = 0$ (the polar axis), since that's the direction the curve approaches the origin from.

16 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 1.01 Integration by Parts

Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.

$\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — these are the four tangent-line directions at the pole for this 4-petaled rose.

17 [Challenging] • Target: Classifying polar curves by equation type • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

a $r = 5$ Circle centered at the pole

b $r = 3 + 3\cos\theta$ Cardioid (coefficient equals constant)

c $r = 1 + 3\sin\theta$ Limaçon with an inner loop (constant $<$ coefficient)

d $r = 4\sin(5\theta)$ Rose with 5 petals (odd $n = 5$)

(a) Circle centered at the pole; (b) Cardioid (coefficient equals constant); (c) Limaçon with an inner loop (constant $<$ coefficient); (d) Rose with 5 petals (odd $n = 5$)

18 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.

If n is odd, the rose has exactly n petals (traced over $[0, \pi]$). If n is even, the rose has $2n$ petals (requires the full $[0, 2\pi]$ to trace, since each petal is retraced with opposite sign after π).

19 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 3.01 Parametric Equations

How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?

$r = 2\sin(4\theta)$: $n = 4$ (even) $\Rightarrow 2(4) = 8$ petals. $r = 5\cos(7\theta)$: $n = 7$ (odd) $\Rightarrow 7$ petals.

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20 [Challenging] • Target: Graphing polar curves point-by-point • Cross-Lesson: 3.02 Vector-Valued Functions

Complete a short table of (θ, r) values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

$\theta = 0$: $r = 4$. $\theta = \frac{\pi}{2}$: $r = 2$. $\theta = \pi$: $r = 0$. $\theta = \frac{3\pi}{2}$: $r = 2$. The curve is widest ($r = 4$) along the polar axis, pinches to a point ($r = 0$) at $\theta = \pi$ (opposite direction), forming the characteristic heart shape.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between polar and rectangular coordinates

How many petals does the rose $r = \cos(2\theta)$ have?

a. 4 ✓ **b.** 6

c. 7 **d.** 2

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

22 [Foundational] • Target: Identifying and sketching common polar curves

How many petals does the rose $r = 4\cos(7\theta)$ have?

a. 14 **b.** 9

c. 7 ✓ **d.** 5

Here $n = 7$, which is odd, so the rose has exactly n petals: 7 petals.

23 [Foundational] • Target: The slope of a polar curve

How many petals does the rose $r = 6\cos(2\theta)$ have?

a. 7 **b.** 4 ✓

c. 2 **d.** 6

Here $n = 2$, which is even, so the rose has $2n$ petals: 4 petals.

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24 [Foundational] • Target: Finding horizontal and vertical tangents

How many petals does the rose $r = 6\sin(7\theta)$ have?

a. 14

b. 5

c. 7 ✓

d. 9

Here $n = 7$, which is odd, so the rose has exactly n petals: 7 petals.

25 [Foundational] • Target: Symmetry of polar curves

How many petals does the rose $r = 2\sin(3\theta)$ have?

a. 6

b. 5

c. 1

d. 3 ✓

Here $n = 3$, which is odd, so the rose has exactly n petals: 3 petals.

26 [Foundational] • Target: More polar-to-rectangular conversions

How many petals does the rose $r = 2\cos(4\theta)$ have?

a. 6

b. 4

c. 10

d. 8 ✓

Here $n = 4$, which is even, so the rose has $2n$ petals: 8 petals.

27 [Foundational] • Target: More slope-of-a-polar-curve practice

How many petals does the rose $r = 2\sin(6\theta)$ have?

a. 14

b. 12 ✓

c. 10

d. 6

Here $n = 6$, which is even, so the rose has $2n$ petals: 12 petals.

28 [Foundational] • Target: Tangent lines at the pole

Convert the polar point $\left(-8, \frac{3\pi}{2}\right)$ to rectangular coordinates.

a. $(-1, 10)$

b. $(8, 0)$

c. $(0, 8)$ ✓

d. $(0, -8)$

$x = r\cos\theta = -8\cos\left(\frac{3\pi}{2}\right) = 0$. $y = r\sin\theta = -8\sin\left(\frac{3\pi}{2}\right) = 8$. Point: $(0, 8)$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Classifying polar curves by equation type**

Convert the polar point $(2, \frac{\pi}{2})$ to rectangular coordinates.

a. $(2, 0)$

b. $(0, 2)$ ✓

c. $(0, -2)$

d. $(2, 4)$

$x = r\cos\theta = 2\cos\left(\frac{\pi}{2}\right) = 0$. $y = r\sin\theta = 2\sin\left(\frac{\pi}{2}\right) = 2$. Point: $(0, 2)$.

30 **[Intermediate]** • **Target: Number of petals in rose curves**

Convert the polar point $\left(-5, \frac{3\pi}{2}\right)$ to rectangular coordinates.

a. $(5, 0)$

b. $(0, 5)$ ✓

c. $(0, -5)$

d. $(2, 3)$

$x = r\cos\theta = -5\cos\left(\frac{3\pi}{2}\right) = 0$. $y = r\sin\theta = -5\sin\left(\frac{3\pi}{2}\right) = 5$. Point: $(0, 5)$.

31 **[Intermediate]** • **Target: Graphing polar curves point-by-point**

Convert the polar point $(2, \pi)$ to rectangular coordinates.

a. $(2, 0)$

b. $(-2, 0)$ ✓

c. $(-1, 0)$

d. $(0, -2)$

$x = r\cos\theta = 2\cos(\pi) = -2$. $y = r\sin\theta = 2\sin(\pi) = 0$. Point: $(-2, 0)$.

32 **[Intermediate]** • **Target: Concavity considerations for polar curves**

Convert the polar point $(-3, 0)$ to rectangular coordinates.

a. $(-3, 0)$ ✓

b. $(3, 0)$

c. $(-2, 0)$

d. $(0, -3)$

$x = r\cos\theta = -3\cos(0) = -3$. $y = r\sin\theta = -3\sin(0) = 0$. Point: $(-3, 0)$.

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33 [Intermediate] • Target: Slopes of Tangent Lines to Polar Curves

Convert the polar point $(9, \frac{\pi}{2})$ to rectangular coordinates.

a. $(0, 9)$ ✓

b. $(0, -9)$

c. $(9, 0)$

d. $(2, 11)$

$x = r\cos\theta = 9\cos\left(\frac{\pi}{2}\right) = 0$. $y = r\sin\theta = 9\sin\left(\frac{\pi}{2}\right) = 9$. Point: $(0, 9)$.

34 [Intermediate] • Target: Horizontal and Vertical Tangents on Polar Curves

Convert the polar point $(-5, \frac{\pi}{2})$ to rectangular coordinates.

a. $(-5, 0)$

b. $(0, -5)$ ✓

c. $(0, 5)$

d. $(2, -3)$

$x = r\cos\theta = -5\cos\left(\frac{\pi}{2}\right) = 0$. $y = r\sin\theta = -5\sin\left(\frac{\pi}{2}\right) = -5$. Point: $(0, -5)$.

35 [Intermediate] • Target: Rate of Change of r and Distance from Origin

Classify the curve $r = 9 + 9\cos\theta$.

a. Circle through the pole

b. Limaçon with an inner loop

c. Limaçon without an inner loop

d. **Cardioid** ✓

For $r = 9 + 9\cos\theta$: the constant term equals the coefficient, the defining property of a cardioid. This is a cardioid.

36 [Intermediate] • Target: Arc Length of Polar Curves

Classify the curve $r = 5 + 5\sin\theta$.

a. Limaçon without an inner loop

b. **Cardioid** ✓

c. Circle through the pole

d. Limaçon with an inner loop

For $r = 5 + 5\sin\theta$: the constant term equals the coefficient, the defining property of a cardioid. This is a cardioid.

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37 [Intermediate] • Target: Converting between polar and rectangular coordinates

Find the smallest positive θ where the rose $r = 4\cos(6\theta)$ passes through the pole.

a. $\frac{7\pi}{12}$

b. $\frac{\pi}{24}$

c. $\frac{\pi}{6}$

d. $\frac{\pi}{12}$ ✓

Set $4\cos(6\theta) = 0 \Rightarrow \cos(6\theta) = 0$. The smallest positive solution is $6\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{12}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Identifying and sketching common polar curves • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the smallest positive θ where the rose $r = 9\cos(4\theta)$ passes through the pole.

a. $\frac{\pi}{16}$

b. $\frac{\pi}{4}$

c. $\frac{5\pi}{8}$

d. $\frac{\pi}{8}$ ✓

Set $9\cos(4\theta) = 0 \Rightarrow \cos(4\theta) = 0$. The smallest positive solution is $4\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{8}$.

39 [Challenging] • Target: The slope of a polar curve • Cross-Lesson: 3.01 Parametric Equations

The limaçon $r = 7 + 14\sin\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{7\pi}{6}$ ✓

b. $\frac{11\pi}{6}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{2}$

Set $r = 0$: $7 + 14\sin\theta = 0 \Rightarrow \sin\theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{7\pi}{6}$.

40 [Challenging] • Target: Finding horizontal and vertical tangents • Cross-Lesson: 3.02 Vector-Valued Functions

The limaçon $r = 8 + 16\sin\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{5\pi}{6}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$ ✓

d. $\frac{11\pi}{6}$

Set $r = 0$: $8 + 16\sin\theta = 0 \Rightarrow \sin\theta = -\frac{1}{2}$. The smaller positive solution is $\theta = \frac{7\pi}{6}$.

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41 [Challenging] • Target: Slopes of Tangent Lines to Polar Curves • Cross-Lesson: 4.02 Area in Polar Coordinates

Which formula gives the slope $\frac{dy}{dx}$ of the tangent line to the polar curve $r = f(\theta)$?

a. $\frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$ ✓

b. $\frac{dr}{d\theta}$

c. $\frac{r\sin\theta}{r\cos\theta}$

d. $\frac{\frac{dr}{d\theta}\cos\theta}{\frac{dr}{d\theta}\sin\theta}$

From $x = r\cos\theta$ and $y = r\sin\theta$, applying product rule: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$.

42 [Challenging] • Target: Slopes of Tangent Lines to Polar Curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

What is the slope of the tangent line to the cardioid $r = 1 + \sin\theta$ at $\theta = 0$?

a. 1 ✓

b. -1

c. 0

d. Undefined

At $\theta = 0$: $r = 1$. $r' = \cos(0) = 1$. Numerator $= 1(0) + 1(1) = 1$. Denominator $= 1(1) - 1(0) = 1$. Slope $= \frac{1}{1} = 1$.

43 [Challenging] • Target: Horizontal and Vertical Tangents on Polar Curves • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

At the pole ($r = 0$), what is the slope of the tangent line to the curve $r = f(\theta)$ when $f(\alpha) = 0$?

a. $\tan\alpha$ ✓

b. $\cot\alpha$

c. 0

d. Undefined

When $r = 0$, $\frac{dy}{dx} = \frac{r'\sin\alpha + 0}{r'\cos\alpha - 0} = \frac{\sin\alpha}{\cos\alpha} = \tan\alpha$.

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44 [Challenging] • Target: Rate of Change of r and Distance from Origin • Cross-Lesson: 6.01 Power Series and Interval of Convergence

If $\frac{dr}{d\theta} > 0$ for a polar curve, what does this indicate about the distance from the pole?

a. The distance from the origin is increasing as θ increases ✓

b. The distance is decreasing

c. The curve is concave up

d. The curve is a line

Since r measures directed distance from the origin, $\frac{dr}{d\theta} > 0$ (for $r > 0$) means the particle is moving away from the pole.

45 [Challenging] • Target: Arc Length of Polar Curves • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Which integral represents the arc length of the polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$?

a. $\int_{\alpha}^{\beta} \sqrt{r^2 + (\frac{dr}{d\theta})^2}\, d\theta$ ✓

b. $\int_{\alpha}^{\beta} \sqrt{1 + (\frac{dr}{d\theta})^2}\, d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$

d. $\int_{\alpha}^{\beta} r d\theta$

By transforming $ds = \sqrt{(dx)^2 + (dy)^2}$, polar arc length is $L = \int_{\alpha}^{\beta} \sqrt{r^2 + (r')^2}\, d\theta$.

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